

**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**

**Regular and Supplementary Summer 2024**

**Course: B. Tech.**

**Semester: VI**

**Branch: Computer Engineering / Computer Science and Engineering**

**Subject Code & Name: BTHM605A\_Y23 Development Engineering**

**Max Marks: 60**

**Date: 25/06/2024**

**Duration: 3.00 Hr.**

**Instructions to the Students:**

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

Mark

**Q. 1 Solve Any Two of the following.**

- |                                                                                                                                                             |   |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| A) What is sustainable development? How does development engineering integrate with sustainable development goals?                                          | 6 |
| B) Define development engineering with the following perspectives:<br>1. Problem-Solving Process<br>2. Economic Development<br>3. Technological Empowerment | 6 |
| C) Enlist and explain the guiding principles of economic development.                                                                                       | 6 |

**Q.2 Solve Any Two of the following.**

- |                                                                                                                     |   |
|---------------------------------------------------------------------------------------------------------------------|---|
| A) What are the prime responsibilities of a development engineer who intends to solve societal problems?            | 6 |
| B) Identify the major issues in the economic development of India and how development engineering can address them. | 6 |
| C) Define the poverty line. Which poverty measure will help us to find the causes and consequences of poverty?      | 6 |

**Q. 3 Solve Any Two of the following.**

- |                                                                                                    |   |
|----------------------------------------------------------------------------------------------------|---|
| A) Which guiding principle of economic development will follow the secular perspective?            | 6 |
| B) Explain why incorporating social justice in engineering is crucial for sustainable development. | 6 |
| C) Explain the Amartya Sen's idea of social justice.                                               | 6 |

**Q.4 Solve Any Two of the following.**

- A) Enlist and explain any three emerging technologies and their contribution to sustainable development. 6
- B) How do artificial intelligence and machine learning have a positive impact on the health sector? 6
- C) How do technological advancements contribute to a quality education? 6

**Q. 5 Solve Any Two of the following.**

- A) Explain the applications of blockchain technology for societal development. 6
- B) Discuss the importance of teamwork and project management in the context of community development projects. 6
- C) Explain the role of artificial intelligence for humanitarian purposes with suitable examples. 6

**\*\*\* End \*\*\***